

Petras Vestartas

Portfolio

www.vestartas.com



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Personal

Neufeldstrasse 19, 3012 Bern

Switzerland

Born 13/12/1989 · Lithuanian

Languages

English **C1**

German **B1**

French **A1**

COMPETENCES

Open-Source Software

Libraries in C++, Python, Rust and C#, released on PyPI, conda-forge and food4rhino

Timber Construction

Joinery, plates and vaulted floors; structural system for the 1st-place Brussels Sports Tower

Robotics & CNC

Commissioned an ABB 6700 robot with 6.7 m track; 5-axis CNC; raw wood scanning

Architectural Design

DMAA Vienna: Austrian Pavilion, Venice Biennale 2016, and two 1st-place competitions

Teaching & Funding

EPFL courses; workshops at AA Visiting School and Zaha Hadid Architects; robotics and timber grants

Research & Publications

Conference and journal papers on timber, robotics and geometry

EXPERIENCE

- 2023-now** **Post-doc** · Block Research Group (BRG), ETH Zurich
Develops the COMPAS framework, with C++ and Fortran90 bindings for Python · researches timber vaulted floors · prepares workshops, developer and industry days · brg.ethz.ch
- 2021-2023** **Post-doc** · Laboratory for Timber Constructions (IBOIS), EPFL
Developed the structural system for the Brussels Sports Tower, 1st-place competition entry · commissioned the ABB 6700 robot and track · developed `compas_wood` · supervised a PhD thesis · epfl.ch/labs/ibois · dfab.ch
- 2016-2017** **Research Assistant** · CITA, Royal Danish Academy of Fine Arts (KADK), Copenhagen
Flora Robotica · Complex Modelling 5 – Inflated Restraint · royaldanishacademy.com/CITA
- 2014-2016** **Architect** · DMAA, Vienna · dmaa.at
Commission Austrian Pavilion, Venice Architecture Biennale 2016
1st place Wohnen am Schweizer Garten, Vienna · Campus Tower, HafenCity Hamburg
3rd place Future Art Lab, Vienna
Recognition Elbbrücken, HafenCity Hamburg
Competitions Porsche Design Tower, Frankfurt · Fiducia GAD, Karlsruhe · Zollhafen, Mainz · U5 Wiener Linien · Taopu and Changchun, China
- 2012-2013** **Architect** · Do-Architects, Vilnius · Vilnius Bajorai masterplan · doarchitects.lt
- 2011** **Intern** · CEBRA, Aarhus · Valer Church · Hoje Taastrup masterplan · cebraarchitecture.dk
- 2010** **Architect** · Vakarinis Fasadas, Siauliai · Palanga Sanatorium · Druskininkai cafeteria
- 2009** **Furniture Designer** · Graniteka, Siauliai · 3D preparation of stone furniture for fabrication

EDUCATION	2017-2021	Doctor of Philosophy (Ph.D.) in Architecture , Laboratory for Timber Construction (IBOIS), École Polytechnique Fédérale de Lausanne (EPFL), Switzerland.	
	2013-2014	Erasmus Exchange Program , Royal Danish Academy of Fine Arts (KADK), Copenhagen, Denmark	
	2012-2014	Master of Science (M.Sc.) in Architecture , Vilnius Academy of Arts (VAA), Lithuania	
	2008-2012	Bachelor of Science with distinction (B.Sc.) in Architecture , Vilnius Academy of Arts (VAA), Lithuania	
SOFTWARE	Code	C++ CGAL, OpenCASCADE, libigl, Eigen, Boost, Open3D, PMP, ShapeOp, OpenGL Python nanobind and pybind11 bindings · COMPAS, Rhino3D · PyPI and conda-forge Rust own geometry kernel · WebGPU (WGPU) C# Rhino and Grasshopper plugins · professional and teaching use Other Fortran90 (LMGC90) · JavaScript (Vue.js) · Java (Processing) · Protobuf · CMake · Git CAD Rhino3D / Grasshopper, Revit, AutoCAD, Adobe, ArcGIS, V-Ray	
	Own tools	<u>session</u> C++ Python Rust one geometry kernel, three languages: same API, same tests <u>wood / compas_wood</u> C++ Python finds where plates touch, cuts fitting joints into both <u>OpenNest</u> C# C++ · <u>compas_nest</u> C++ Python 2D nesting for CNC and laser cutting <u>NGon</u> C# polygonal mesh processing for Grasshopper	
	COMPAS	<u>compas_cg</u> C++ Python CGAL bindings on PyPI and conda-forge <u>compas_occt</u> C++ Python OpenCASCADE BRep and NURBS backend <u>compas_libigl</u> C++ Python · <u>compas_shapeop</u> C++ Python · <u>compas</u> Python <u>compas_nanobind_package_template</u> C++ Python template for C++ extensions	
	BRG	<u>compas_model</u> Python one model for design, analysis and fabrication <u>RhinoVAULT</u> Python · <u>compas_grid</u> Python · <u>compas_cra</u> Python <u>compas_imgc90</u> Fortran90 C++ Python LMG90 discrete element solver bindings <u>compas_manifold</u> C++ Python guaranteed-manifold mesh booleans	
	IBOIS	<u>Cockroach</u> C++ C# point clouds with CGAL, Open3D and CILANTRO <u>COMPAS_ABB6700_IRBT</u> Python ROS ABB 6700 robot on a linear track <u>Raccoon</u> C# 5-axis CNC G-code	
	Rhino	<u>Fox</u> C# · <u>Scatter</u> C# · <u>Mesh Curvature</u> C# · <u>Boundary First Flattening</u> C#	
AWARDS · GRANTS	2026	ETHZ D-ARCH Enabling Grant · to prepare larger funding bids such as Innosuisse	CHF 15'000
	2023	ETHZ Partnership Council for Sustainable Construction and Digital Fabrication · Timber Vaulted Floor materials and CNC fabrication	CHF 19'000
	2022	ENAC Equipment Call, EPFL · high-precision scanners, robotic integration parts	CHF 33'949
	2022	Best Paper Award – Runner-Up · CAADRIA, for the Cockroach point-cloud paper	
	2020-2022	EPFL Equipment Funding · ABB 6700 robot and 6.7 m track · specifications with integrators and suppliers · spindle, tool-changer and multi-move documents	CHF 182'240 + CHF 45'000
	2013	Lithuanian President Antanas Smetona Scholarship · one year, for excellent grades	
TEACHING	2026	Workshop · HSLU · new compas_wood version · advances in timber construction	
	2025	AA Visiting School Seoul · COMPAS Python framework courses	
	2025	Workshop · Zaha Hadid Architects, London · C++ / Python bindings for developers	
	2025	Workshop · Calgary, Canada · COMPAS basics, RhinoVAULT form finding, fabrication	
	2025	Support team · AEC Hackathon · COMPAS installation and new features	

2024 **Workshops** · IBOIS, EPFL · IASS 2024 · compas_wood joinery · BIM with COMPAS to IFC

2023-2025 **Course assistant** · Computational Structural Design (CSD I) · graphic statics with COMPAS AGS

2020-2022 **Co-taught course** · EPFL · [Intro to Computational Architecture \(AR-327\)](#)

2020-2021 **Master's thesis supervision** · EPFL · Maxim Andrist: theory, project, robotic cutting · [thesis record](#)

2020-2021 **ENAC Week** · IBOIS + ALICE, EPFL · "Land of Thousand Dance" · reused timber

2017-2022 **Studio teaching** · EPFL · MA1, MA2 and BA/MA studios · weekly project reviews

2017-2022 **CNC & robot training** · IBOIS, EPFL · Raccoon · Maka 5-axis CNC · ABB robots

2017, 2019 **Workshops** · VAA, Vilnius · graph-based aggregations · 1:1 corrugated cardboard shell

Online **Teaching resource** · EPFL · [Advanced Timber Plate Structural Design](#) (timber joinery generation)

PUBLICATIONS

2025 **Vestartas P.**, Füssler L., Rad A. R., Lee D. S., van Mele T., Block P. **Exploring the Potential of Funicular Timber Floors**. ICSA, Antwerp.

2022 Settimi A., **Vestartas P.**, Gamero J., Weinand Y. **Cockroach: an open-source tool for point cloud processing in CAD**. CAADRIA, Sydney. Best Paper Award – Runner-Up.

Rogean N., Rezaei Rad A., **Vestartas P.**, Latteur P., Weinand Y. **A Collaborative Workflow to Automate the Design, Analysis, and Construction of Integrally-Attached Timber Plate Structures**. CAADRIA, Sydney.

2021 **Vestartas P.**, Rezaei Rad A., Weinand Y. **Robotically-Fabricated Nexorades from Whole Timber**. fib Symposium on the Conceptual Design of Structures, Switzerland.

Rezaei Rad A., Burton H., Rogean N., **Vestartas P.**, Weinand Y. **A framework to automate the design of digitally-fabricated timber plate structures**. Computers & Structures.

2020 **Vestartas P.**, Weinand Y. **Joinery Solver for Whole Timber Structures**. WCTE 2020, Santiago.

Vestartas P., Weinand Y. **Laser Scanning with Industrial Robot Arm for raw wood Fabrication**. ISARC 2020, Kitakyushu, pp. 773–780.

Vestarte L., **Vestartas P.**, Kucinskas R. **Corrugated Cardboard Shell: A Pavilion Project of An Architectural Workshop**. Advances in Architectural Geometry (AAG).

2019 **Vestartas P.**, Rogean N., Gamero J., Weinand Y. **Modelling Workflow for Segmented Timber Shells using Wood-wood Connections**. Design Modelling Symposium, Berlin, pp. 596–607.

Vestartas P., Palletier L., Nakad M. T., Rad A. R., Weinand Y. **Segmented Spiral Using Inter-Connected Timber Elements**. IASS Symposium, Barcelona.

Nguyen A. C., Himmer B., **Vestartas P.**, Weinand Y. **Performance Assessment of Double-Layered Timber Plate Shells using Alternative Structural Systems**. IASS Annual Symposium.

Nguyen A. C., **Vestartas P.**, Weinand Y. **Design framework for the structural analysis of free-form timber plate structures using wood-wood connections**. Automation in Construction.

2017 Ayres P., **Vestartas P.**, Thomsen M. R. **Enlisting Clustering and Graph-Traversal Methods for Cutting Pattern and Net Topology Design in Pneumatic Hybrids**. Design Modelling Symposium, Paris.

Vestartas P., Heinrich M. K., Zwierzycki M., Leon D. A., Cheheltan A., La Magna R., Ayres P. **Design Tools and Workflows for Braided Structures**. Design Modelling Symposium, Paris.

Zwierzycki M., **Vestartas P.**, Heinrich M. K., Ayres P. **High Resolution Representation and Simulation of Braiding Patterns**. ACADIA 2017.

Books

2022 · **Les cahiers de l'Ibois 3 / Ibois Notebooks 3**. Introductory text and images, with Berthier S. et al.

2021 · **Structural design methodology in Integrally-Attached Timber Plate structures**. With Rezaei Rad A., in Design of Integrally-Attached Timber Plate Structures, Routledge.

2020 · R. Kucinskas (ed.), **Digital Architecture at the VAA** · 2016 · D. Kohler, **The Mereological City** · 2014 · **Best graduation projects of architecture students in Baltic states**